

Grade 6 ICT Notes – Let's Get to Know the Computer

1. What is a Computer?

A **computer** is an electronic device that helps people perform many tasks quickly and accurately. It can work for a long time without getting tired and can process large amounts of information.

Advantages of a Computer

- Works very fast.
- Gives accurate results when correct data and instructions are provided.
- Does not get tired.
- Can perform the same task repeatedly with consistent results.
- Can store a large amount of data.
- Retrieves stored information whenever needed.
- Follows instructions exactly.
- Cannot think or make decisions like humans.
- High initial cost but relatively low maintenance cost.

2. Basic Functions of a Computer

A computer performs three main functions.

Input → Process → Output

Input

Entering data and instructions into the computer.

Examples of input devices:

- Keyboard
- Mouse
- Scanner
- Microphone
- Webcam
- Barcode Reader

Process

The computer processes the entered data according to the given instructions.

Output

The processed information is displayed or produced.

Examples of output devices:

- Monitor
- Printer
- Speaker
- Multimedia Projector

3. Characteristics of a Computer

- **Speed** – Can perform billions of operations in one second.
- **Accuracy** – Produces correct results if correct data and instructions are given.
- **Reliability** – Gives consistent output for the same input.
- **Storage** – Can store a large amount of information.
- **Automation** – Works automatically according to instructions.
- **No Intelligence** – Cannot think or make decisions independently.

4. Embedded Computer Systems

An **Embedded Computer System** is a computer program built into a device to perform a specific function automatically.

Examples

- Smart Phones
- Washing Machines
- Modern Cars
- Smart Televisions
- Digital Thermometers

5. Components of a Computer

A computer consists of several important components.

a) Input Devices

Used to enter data and instructions into the computer.

Examples

- Keyboard
- Mouse
- Scanner
- Microphone
- Webcam
- Barcode Reader

b) Output Devices

Used to display or produce processed information.

Examples

- Monitor
- Printer
- Speaker
- Multimedia Projector

c) Central Processing Unit (CPU)

- Known as the **brain of the computer**.
- Controls all computer operations.
- Processes data according to instructions.
- Located inside the **System Unit** on the **Motherboard**.

d) Main Memory (RAM)

- RAM stands for **Random Access Memory**.
- Temporarily stores data, instructions, and information while the computer is working.
- Data stored in RAM is lost when the computer is turned off.

e) Storage Devices

Used to store data permanently.

Internal Storage

- Hard Disk Drive (HDD)

External Storage

- External Hard Disk
- Pen Drive (USB Flash Drive)
- CD
- DVD
- Blu-ray Disc

f) Communication Devices

Used to exchange data between computers and networks.

Examples

- Modem
- Router
- Network Interface Card (NIC)

6. Hardware and Software

Hardware

The physical parts of a computer that can be seen and touched.

Examples

- Monitor

- Keyboard
- Mouse
- CPU
- Printer

Software

A collection of programs that tells the computer what to do.

Examples

- Microsoft Word
- Paint
- Media Player
- Computer Games
- Video Editing Software

User

A **user** is the person who operates a computer.

7. Applications of Computers

Computers are used in many areas of daily life.

a) Schools

- Teaching and learning
- Presentations
- Internet research
- Maintaining student records
- Preparing reports

b) Banks

- ATM services
- Internet Banking
- Mobile Banking
- Electronic card payments
- Managing customer accounts

c) Hospitals

- Digital thermometers
- ICU monitoring systems
- Operation theatre equipment
- Maintaining patient records

d) Factories

- Robotic technology
- Automated production
- Increased productivity
- Reduced human labour

e) Agriculture

- Automatic watering systems
- Fertilizer distribution
- Harvesting machines
- Weed control
- Increased production with lower costs